

```
import pandas as pd
import numpy as np
```

```
data=pd.read_csv('/content/data.csv')
```

data



	ad_id	type	price	area	bedrooms	bathrooms	level	furnished	rent	city	region
0	Ad id 194281336	apartment	3790000.0	207.0	3	3	2	NaN	no	giza	6th of october
1	Ad id 194198684	apartment	870000.0	155.0	3	3	4	no	no	cairo	new heliopolis
2	Ad id 191102186	apartment	1546400.0	170.0	3	2	9	NaN	no	cairo	zahraa al maadi
3	Ad id 193951860	apartment	1300000.0	270.0	4	3	ground	no	no	sharqia	10th of ramadan
4	Ad id 194064010	duplex	5888000.0	387.0	4	4	ground	no	no	cairo	new cairo - el tagamoa
...	...	...	...	...	...	...	...	...	...	...	...
54324	Ad id 193833696	apartment	18000.0	220.0	2	3	2	yes	yes	alexandria	kafr abdo
54325	Ad id 193833574	apartment	200.0	70.0	1	1	1	yes	yes	alexandria	mandara
54326	Ad id 186376403	apartment	2500.0	80.0	2	1	ground	no	yes	alexandria	miami
54327	Ad id 193831871	apartment	14000.0	200.0	3	2	10+	yes	yes	alexandria	al ibrahimiyyah
54328	Ad id 193831851	apartment	450.0	120.0	2	1	10+	yes	yes	alexandria	mandara

Next steps: [Generate code with data](#) [View recommended plots](#) [New interactive sheet](#)

data.info()



```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 54329 entries, 0 to 54328
Data columns (total 11 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   ad_id       54329 non-null  object
1   type        54329 non-null  object
2   price       54329 non-null  float64
3   area        54329 non-null  float64
4   bedrooms    54329 non-null  object
5   bathrooms   54329 non-null  object
6   level       54329 non-null  object
7   furnished   42448 non-null  object
8   rent        54329 non-null  object
9   city        54329 non-null  object
10  region      54329 non-null  object
dtypes: float64(2), object(9)
memory usage: 4.6+ MB
```

data.isnull().sum()



	0
ad_id	0
type	0
price	0
area	0
bedrooms	0
bathrooms	0
level	0
furnished	11881
rent	0
city	0
region	0

```
duplicates = data[data.duplicated()]
duplicates
```

	ad_id	type	price	area	bedrooms	bathrooms	level	furnished	rent	city	region
15619	Ad id 194199658	apartment	780000.0	130.0	3	2	3	no	no	giza	hadayek al-ahram
15620	Ad id 194304726	apartment	4000000.0	171.0	3	2	7	no	no	alexandria	smoha
15621	Ad id 194315592	apartment	2900000.0	137.0	2	2	2	no	no	cairo	mostakbal city
15622	Ad id 194289013	apartment	2885200.0	147.0	2	2	5	no	no	cairo	new cairo - el tagamoa
15623	Ad id 192153664	apartment	2400000.0	197.0	3	2	1	no	no	cairo	new cairo - el tagamoa
...	...	...	...	...	...	...	...	...	...	...	...
53650	Ad id 194186535	apartment	2750000.0	135.0	2	1	5	yes	no	alexandria	glim
53654	Ad id 194187746	apartment	2850000.0	160.0	2	2	3	yes	no	alexandria	miami
53658	Ad id 194187474	apartment	2080000.0	109.0	2	2	3	no	no	alexandria	smoha
53716	Ad id 191302240	apartment	2850000.0	320.0	4	2	3	NaN	no	alexandria	laurent
54211	Ad id 193866800	apartment	5500.0	90.0	2	1	6	yes	yes	alexandria	kafr abdo

Next steps: [Generate code with duplicates](#) [View recommended plots](#) [New interactive sheet](#)

```
duplicate=data.duplicated()
duplicate.sum()
```

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```
!pip install sweetviz
!pip install pandas
!pip install dataprofiler
!pip install ydata_profiling
```



```
Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from pydantic>=2->ydata_profiling)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.24.0->ydata_profiling)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.24.0->ydata_profiling)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.24.0->ydata_profiling)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.24.0->ydata_profiling)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.11/dist-packages (from statsmodels<1,>=0.13.2->ydata_profiling)
Requirement already satisfied: attrs>=19.3.0 in /usr/local/lib/python3.11/dist-packages (from visions<0.8.2,>=0.7.5->visions[type_inspection])
Requirement already satisfied: networkx>=2.4 in /usr/local/lib/python3.11/dist-packages (from visions<0.8.2,>=0.7.5->visions[type_inspection])
Requirement already satisfied: puremagic in /usr/local/lib/python3.11/dist-packages (from visions<0.8.2,>=0.7.5->visions[type_inspection])
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.7->matplotlib<=3.10
```

```
import sweetviz as sv
import pandas as pd
```

```
report = sv.analyze(data)
report.show_html('sweetviz_report.html')
```

Done! Use 'show' commands to display/save. [100%] 00:00 -> (00:00 left)

```
from ydata_profiling import ProfileReport
import pandas as pd
```

```
profile = ProfileReport(data, title="Data Profiling Report")
profile.to_file("report.html")
```

[Upgrade to ydata-sdk](#)

Improve your data and profiling with ydata-sdk, featuring data quality scoring, redundancy detection, outlier identification, text validation, and synthetic data generation.

Summarize dataset: 100% 24/24 [00:03<00:00, 6.83it/s, Completed]

0%		0/11 [00:00<?, ?it/s]
9%		1/11 [00:00<00:06, 1.57it/s]
36%		4/11 [00:00<00:01, 6.56it/s]
55%		6/11 [00:01<00:00, 7.27it/s]
73%		8/11 [00:01<00:00, 6.27it/s]
100%		11/11 [00:01<00:00, 7.01it/s]

Generate report structure: 100% 1/1 [00:05<00:00, 5.39s/it]

Render HTML: 100% 1/1 [00:00<00:00, 1.51it/s]

```
import json
from dataprofiler import Data, Profiler
# Profile the dataset
profile = Profiler(data) # Calculate Statistics, Entity Recognition, etc
```

```
# Generate a report and use json to prettify.
report = profile.report(report_options={"output_format":"pretty"})
```

```
# Print the report
print(json.dumps(report, indent=4))
```



```
    "IPV6": 0.0,  
    "MAC_ADDRESS": 0.0,  
    "PERSON": 0.001,  
    "PHONE_NUMBER": 0.0,  
    "SSN": 0.0,  
    "URL": 0.0,  
    "US_STATE": 0.0,  
    "INTEGER": 0.0,  
    "FLOAT": 0.0,  
    "QUANTITY": 0.0,  
    "ORDINAL": 0.0  
  },  
  "sample_size": 10865,  
  "null_count": 0,  
  "null_types": [],  
  "null_types_index": {},  
  "data_type_representation": {  
    "datetime": 0.0,  
    "int": 0.0,  
    "float": 0.0,  
    "string": 1.0  
  }  
}  
}  
}  
}
```

Start coding or [generate](#) with AI.